Assignment: 4

Based on Revisit C programming (By Dr. Ranjit Rajak)

Write output with detailed justifications

1. #include <stdio.h>

```
int main()
{
printf("%d", printf("Hello World!"));
return 0;
}
```

2. #include <stdio.h>

```
int main() {
    int x = 10, y = 20;
    x = x + y - (y = x);
    printf("%d %d", x, y);
    return 0;
}
```

3. #include <stdio.h>

```
int main() {
    int num = 1;
    if (num = 0)
    printf("The number is 0");
    else
    printf("The number is not 0");
    return 0;
}
```

4. #include <stdio.h>

```
int main() {
    char *ptr = "Hello World";
    printf("%c", *&*ptr);
    return 0;
}
```

5.
```
#include <stdio.h>

int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int *ptr = arr;
    printf("%d", *++ptr);
    return 0;
}
```

6.
```
#include <stdio.h>

int main() {
    printf("%d", sizeof("Hello"));
    return 0;
}
```

7.
```
#include <stdio.h>
int main() {
    int x = 10;
    int y = x++;
    printf("%d %d", x, y);
    return 0;
}
```

8.
```
#include <stdio.h>

void func(int x) {
    x = x + 2;
}
int main() {
    int x = 5;
```

```
    func(x);
    printf("%d", x);
    return 0;
}
```

9.
```
#include <stdio.h>

   int main() {
    int x = 0, y = 0;
    while (++x < 5) y++;
    printf("%d %d", x, y);
    return 0;
    }
```

10.
```
#include <stdio.h>

int main() {
    int x = 5;
    int y = x = 0;
    printf("%d %d", x, y);
    return 0;
}
```

11.
```
#include <stdio.h>

int main() {
    int arr[] = {1, 2, 3, 4, 5};
    int *p = arr;
    ++*p;
    p += 2;
    printf("%d", *p);
    return 0;
}
```

12.
```
#include <stdio.h>

int main() {
```

```
    int arr[2][3] = {{1, 2, 3}, {4, 5, 6}};
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 3; j++) {
            printf("%d ", arr[i][j]);
        }
    }
    return 0;
}
```

13.
```
#include <stdio.h>

int main() {
    char c = 255;
    int i = c;
    printf("%d", i);
    return 0;
}
```

14.
```
#include <stdio.h>

int main() {
    int i = 10;
    do {
        printf("%d ", i);
        i++;
    } while (i < 10);
    return 0;
}
```

15.
```
#include <stdio.h>

 int main() {
    int a = 10, b = 20;
    const int *ptr = &a;
    ptr = &b;
    printf("%d", *ptr);
```

```
    return 0;
}
```

16.
```
#include <stdio.h>

int main() {
    int arr[3] = {0, 1, 2};
    printf("%d", *(arr + 1));
    return 0;
}
```

17.
```
#include <stdio.h>

int main() {
    int a = 5, b = 10, max;
    max = (a > b) ? a : b;
    printf("%d", max);
    return 0;
}
```

18.
```
#include <stdio.h>

int main() {
    int x = 1, y = 2, z = 3;
    printf("%d", x < y < z);
    return 0;
}
```

19.
```
#include <stdio.h>

int main() {
    int nums[] = {5, 10, 15, 20, 25};
    int *ptr = nums + 2;
    printf("%d", *ptr);
    return 0;
}
```

20.
```
#include <stdio.h>
```

```
int main() {
    for (int i = 0; i < 5; i++) {
        if (i == 3) continue;
        printf("%d ", i);
    }
    return 0;
}
```

21. #include <stdio.h>

```
int main() {
    int x = 10, y = 15;
    printf("%d", x & y);
    return 0;
}
```

22. #include <stdio.h>

```
int main() {
    int a = -10;
    unsigned int b = 6;
    printf("%d", a + b);
    return 0;
}
```

23. include <stdio.h>

```
void func() {
    static int count = 0;
    count++;
    printf("%d ", count);
}
int main() {
    func();
    func();
    func();
```

```
    return 0;
}
```

24.
```
#include <stdio.h>

int main() {
    int arr[] = {10, 20, 30, 40};
    printf("%d", 2[arr]);
    return 0;
}
```

25.
```
#include <stdio.h>

int main() {
    int x = 5;
    int y = x++ + ++x;
    printf("%d %d", x, y);
    return 0;
}
```

26.
```
#include <stdio.h>

int main() {
    int arr[10];
    printf("%d", sizeof(arr) / sizeof(arr[0]));
    return 0;
}
```

27.
```
#include <stdio.h>

int main() {
    int x = 5;
    printf("%d %d %d", x, x++, ++x);
    return 0;
}
```

28.
```
#include <stdio.h>
```

```
int main() {
    int arr[3] = {1, 2, 3};
    printf("%d", arr[5]);
    return 0;
}
```

29.
```
#include <stdio.h>

int main() {
    int i = 0;
    while (i < 3)
        printf("Hi\n");
    return 0;
}
```

30.
```
#include <stdio.h>

int main() {
    int x = 0;
    if (x = 5)
        printf("True");
    else
        printf("False");
    return 0;
}
```

31.
```
#include <stdio.h>

int main() {
    float a = 0.1;
    if (a == 0.1)
        printf("Equal");
    else
        printf("Not Equal");
    return 0;
}
```

32.
```
#include <stdio.h>
```

```
int main()
 { int x = (10, 20, 30);
printf("%d", x); return 0; }
```

33. #include <stdio.h> int main() { int x = 0; if(x && printf("Hello")) printf("World"); return 0; }

34. #include <stdio.h> int main() { int x = 10; printf("%lu %d", sizeof(x++), x); return 0; }

35. #include <stdio.h>

```
int main()
 { int a = 5, b = 10;
int result = (a > b) ? a : (b > 15 ? b : 20);
printf("%d", result);
return 0;
}
```

36. #include <stdio.h>

```
int main()
{ int x = 1;
if(x || printf("Hello"))
printf("World");
return 0;
 }
```

37. #include <stdio.h>

```
int main()
```

```
{ int num = 17;
if(num & 1)
printf("Odd");
 else printf("Even");
return 0;
}
38. #include <stdio.h>
int main()
 {
int a = 5, b = 7;
a = a ^ b;
b = a ^ b;
a = a ^ b;
printf("%d %d", a, b);
 return 0;
}
```

Submit on or Before 10th September 2026.